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Database Foundations

6-5

Transaction Control Language (TCL)

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Objectives

- This lesson covers the following objectives:
 - Describe the purpose of the Transaction Control Language (TCL)
 - Explain the TCL operations that are required to manage a transaction:
 - COMMIT
 - SAVEPOINT
 - ROLLBACK
 - Describe the need for read consistency



Application Express

- COMMIT, ROLLBACK, and SAVEPOINT are not supported in Oracle Application Express, due to the way Oracle Application Express manages connections to the database



Oracle APEX

Database Transactions

- Transactions consist of DML statements that represent one consistent change to the data
- The Oracle server ensures data consistency based on transactions
- Transactions give you more flexibility and control when you change data, and they ensure data consistency in the event of user process failure or system failure

Database Transactions

- For example, a transfer of funds between two accounts should include the debit in one account and the credit to another account in the same amount. Both actions should either fail or succeed together; the credit should not be committed without the debit.

Database Transactions

- A database transaction consists of one of the following statements:
 - DML statements that represent one consistent change to the data
 - One DDL statement
 - One TCL statement

Database Transactions: Start and End

- A transaction begins when the first DML SQL statement is executed
- It ends with one of the following events:
 - A COMMIT or ROLLBACK statement is issued
 - A DDL or TCL statement executes (automatic commit)
 - The user exits SQL software being used
 - The system crashes

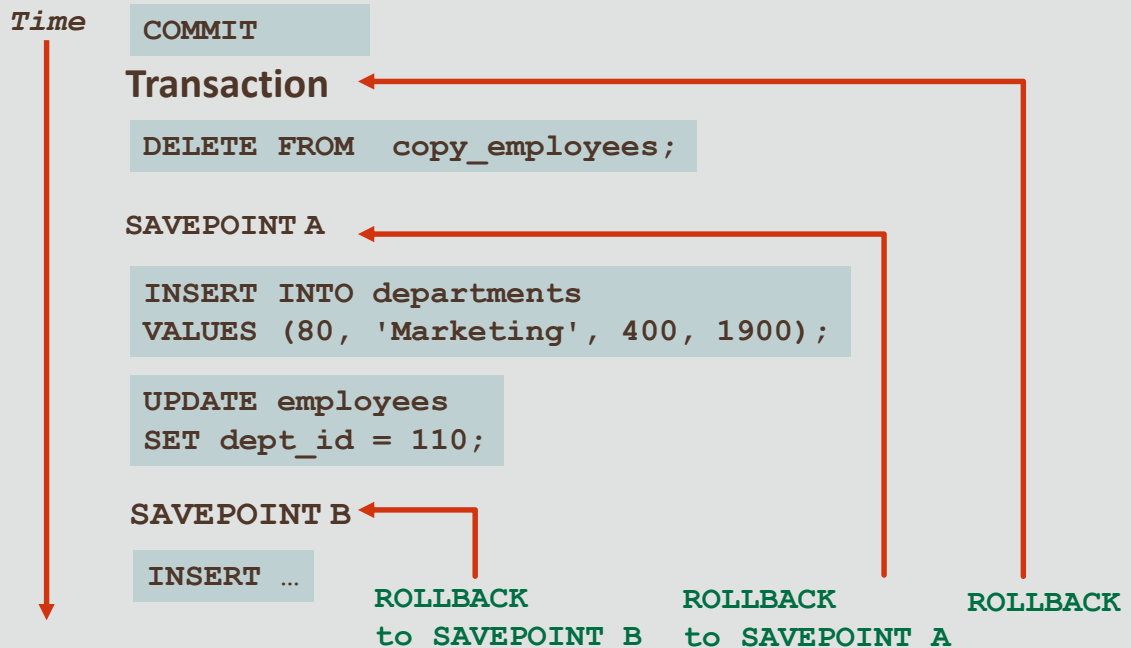
Advantages of COMMIT and ROLLBACK Statements

- With COMMIT and ROLLBACK statements, you can:
 - Ensure data consistency
 - Preview data changes before making changes permanent
 - Group logically related operations
 - Have control over permanent changes to the data

Transaction Control Statements

Statement	Description
COMMIT	Ends the current transaction by making all pending data changes permanent.
SAVEPOINT name	Marks a savepoint within the current transaction.
ROLLBACK	Ends the current transaction by discarding all pending data changes.
ROLLBACK TO SAVEPOINT name	Rolls back the current transaction to the specified savepoint, thereby discarding any changes and/or savepoints that were created after the savepoint to which you are rolling back. If you omit the TO SAVEPOINT clause, the ROLLBACK statement rolls back the entire transaction. Because savepoints are logical, there is no way to list the savepoints that you created.

Explicit Transaction Control Statements



You can control the logic of transactions by using the COMMIT, SAVEPOINT, and ROLLBACK statements.

Note: You cannot COMMIT to a SAVEPOINT. SAVEPOINT is not ANSI-standard SQL.

Note: COMMIT and ROLLBACK are not supported currently in APEX.

Rolling Back Changes to a Marker

- Create a marker in the current transaction by using the `SAVEPOINT` statement
- Discard pending changes by rolling back to that marker by using the `ROLLBACK TO SAVEPOINT` statement

```
UPDATE...  
SAVEPOINT update_done;
```

```
SAVEPOINT update_done succeeded.
```

```
INSERT...  
ROLLBACK TO update_done;
```

```
ROLLBACK TO succeeded.
```

Note: If you create a second savepoint with the same name as an earlier savepoint, the earlier savepoint is deleted.

Implicit Transaction Processing

- An automatic commit occurs under the following circumstances:
 - A DDL statement is issued
 - A TCL statement is issued
 - There is a normal exit from SQL software, without explicitly issuing COMMIT or ROLLBACK statements
- An automatic rollback occurs when there is an abnormal termination of SQL software or when there is a system failure to protect database integrity

State of the Data Before COMMIT or ROLLBACK

- Every data change made during a transaction is temporary if not committed therefore the previous state of the data can be recovered
- The current session can review the results of the DML operations by using the SELECT statement
- Other sessions cannot view the results of the DML statements issued by the current session
- The affected rows are locked; other sessions cannot change the data in the affected rows

State of the Data After COMMIT

- Data changes are saved in the database
- The previous state of the data is overwritten
- All sessions can view the results
- Locks on the affected rows are released; those rows are available for other sessions to manipulate
- All savepoints are erased

Committing Data

- Make the changes:

```
DELETE
FROM   copy_employees
WHERE  employee_id=113;
```

```
1 row(s) deleted.
```

```
INSERT INTO copy_departments
VALUES (290, 'Corporate Tax', NULL, 1700);
```

```
1 row(s) inserted.
```

- Commit the changes:

```
COMMIT;
```

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The following example removes departments 290 and 300 in the copy_departments table, updates a row in the copy_employees table, and saves the data change.

```
DELETE FROM copy_departments
WHERE department_id IN (290, 300);
```

```
UPDATE copy_employees
SET department_id = 80
WHERE employee_id = 206;
```

```
COMMIT;
```

Note: COMMIT is not supported currently in APEX.

State of the Data After ROLLBACK

- Discard all pending changes by using the ROLLBACK statement:
 - Data changes are undone
 - Previous state of the data is restored
 - Locks on the affected rows are released

```
DELETE FROM copy_employees;  
ROLLBACK ;
```

State of the Data After ROLLBACK: Example

- Assume there is a test table containing 4 records:

```
DELETE FROM test;
```

4 rows deleted

```
ROLLBACK;
```

Rollback complete

```
DELETE FROM test WHERE id = 100;
```

1 row deleted

```
SELECT * FROM test WHERE id = 100;
```

No rows selected

```
COMMIT;
```

Commit complete

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Note: ROLLBACK is not supported currently in APEX.

Statement-Level Rollback

- If a single DML statement fails during execution, only that statement is rolled back
- The Oracle server implements an implicit savepoint
- All other changes are retained
- The user should terminate transactions explicitly by executing a COMMIT or a ROLLBACK statement

The Oracle server issues an implicit commit before and after any DDL statement.

So, even if your DDL statement does not execute successfully, you cannot roll back the previous statement because the server issued a commit.

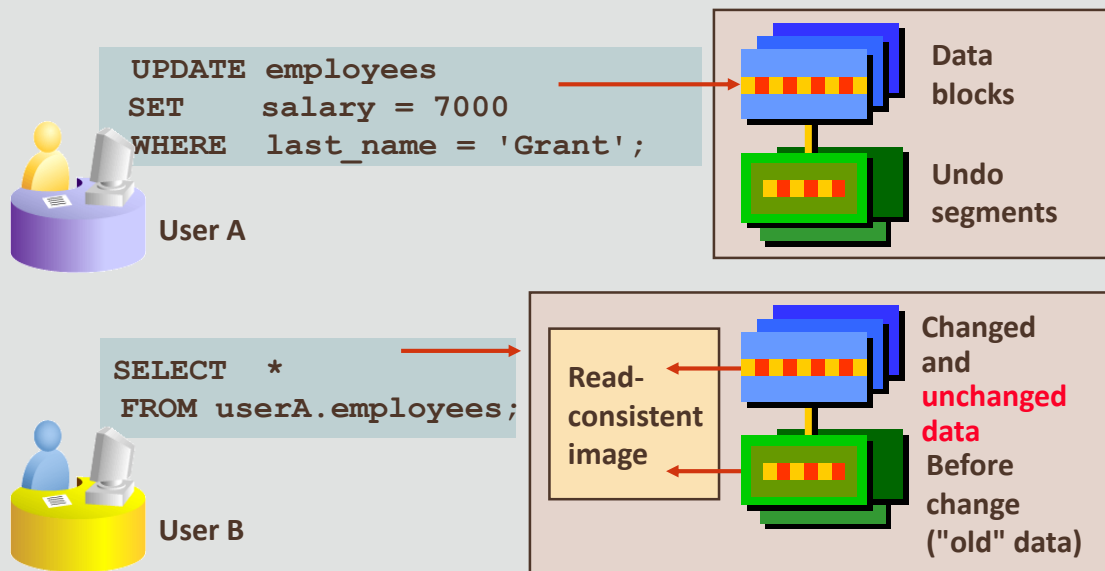
Terminate your transactions explicitly by executing a COMMIT or a ROLLBACK statement.

Read Consistency

- Read consistency guarantees a consistent view of the data at all times – each user sees data as it existed as of the last COMMIT
- Changes made by one user do not conflict with the changes made by another user
- Read consistency ensures that, on the same data:
 - Readers do not wait for writers
 - Writers do not wait for readers
 - Writers wait for other writers

Note: The same user can log in to different sessions. Each session maintains read consistency irrespective of the session where the user logged in.

Implementing Read Consistency



Note : for a more detailed explanation of read consistency see slide notes

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Read consistency is an automatic implementation. It keeps a partial copy of the database in the undo segments. The read-consistent image is constructed from the committed data in the table and the old data that is being changed, and is not yet committed from the undo segment.

When an insert, update, or delete operation is made on the database, the Oracle server takes a copy of the data before it is changed and writes it to an *undo segment*.

All readers, except the one who issued the change, see the database as it existed before the changes started; they view the undo segment's "snapshot" of the data.

Before the changes are committed to the database, only the user who is modifying the data sees the database with the alterations. Everyone else sees the snapshot in the undo segment. This guarantees that readers of the data read consistent data that is not currently undergoing change.

When a DML statement is committed, the change made to the database becomes visible to anyone issuing a SELECT statement after the commit is done. The space occupied by the old data in the undo segment file is freed for reuse.

If the transaction is rolled back, the changes are undone: The original, older version of the data in the undo segment is written back to the table, and all users see the database as it existed before the transaction began.

Summary

- In this lesson, you should have learned how to:
 - Describe the purpose of the Transaction Control Language (TCL)
 - Explain the TCL operations that are required to manage a transaction:
 - COMMIT
 - SAVEPOINT
 - ROLLBACK
 - Describe the need for read consistency



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